

DAYTONA BEACH, FL, USA

WMO: 747870

Lat: 29.183N

Lon: 81.048W

Elev: 31

StdP: 14.68

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 12834

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	35.7	39.6	20.3	15.3	44.0	25.9	19.9	46.5	23.3	61.6	20.4	59.8	5.7	320	0.358

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.6	92.6	77.2	90.9	77.2	89.5	77.1	80.4	87.6	79.6	86.9	78.8	85.9	8.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.5	148.0	83.4	77.6	143.9	82.8	76.9	140.4	82.4	43.8	87.8	42.9	86.8	42.2	86.0	86.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.3	17.9	16.1	30.0	96.1	3.4	1.8	27.6	97.4	25.6	98.4	23.7	99.4	21.2	100.7		
			26.4	81.7	3.5	1.6	23.9	82.9	21.8	83.8	19.9	84.6	17.3	85.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	71.5	58.5	61.4	65.1	70.2	75.7	80.1	81.6	81.8	80.0	74.5	66.8	62.1
	DBStd	10.00	8.55	7.99	7.07	5.40	4.00	2.65	1.98	1.94	2.36	5.43	6.78	8.39
	HDD50	49	25	10	2	0	0	0	0	0	0	0	1	11
	HDD65	709	232	147	87	19	1	0	0	0	0	8	60	155
	CDD50	7913	289	329	471	606	797	903	980	985	900	761	506	386
	CDD65	3095	30	46	90	175	332	453	515	520	450	303	115	66
	CDH74	25849	131	243	512	1162	2704	4006	4964	5105	4036	2209	529	248
	CDH80	8438	6	28	88	245	840	1492	1979	1959	1270	453	60	18
Wind	WSAvg	6.6	6.9	7.0	7.6	7.3	7.0	5.9	5.2	5.4	6.4	7.1	6.8	6.4
Precipitation	PrecAvg	48.5	2.6	3.1	3.5	2.1	3.0	6.3	5.8	6.2	6.6	4.1	2.8	2.4
	PrecMax	74.0	7.2	9.1	10.6	6.2	12.3	15.2	14.4	11.6	16.1	11.7	10.9	12.0
	PrecMin	33.4	0.3	0.4	0.3	0.0	0.1	0.8	0.2	2.0	0.4	0.2	0.0	0.2
	PrecStd	9.8	1.9	2.2	2.7	1.7	2.5	3.7	3.1	2.7	3.5	2.8	2.5	2.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	80.9	83.3	86.1	88.6	93.1	95.5	94.5	93.7	91.5	88.5	84.6	82.2
		MCWB	69.1	69.6	69.4	70.9	72.9	77.1	78.3	77.9	76.5	76.1	73.3	72.0
	2%	DB	77.8	80.2	82.4	85.5	89.8	91.9	92.0	91.6	89.2	86.1	81.7	79.4
		MCWB	67.8	68.1	68.8	70.8	72.7	76.9	78.2	78.1	76.8	75.8	71.9	70.5
	5%	DB	74.7	77.1	79.4	82.7	87.2	89.6	90.3	90.1	87.7	84.3	79.3	76.9
		MCWB	66.4	67.3	67.6	69.8	73.0	76.9	78.1	78.1	76.8	74.6	70.3	69.0
	10%	DB	71.9	73.8	76.5	80.4	84.5	87.6	88.6	88.4	86.3	82.5	77.1	74.1
		MCWB	64.9	65.5	66.1	68.8	72.5	76.4	77.7	77.9	76.4	73.5	69.0	67.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.2	72.6	73.3	75.4	77.5	80.6	81.4	81.9	80.5	79.7	76.5	74.2
		MCDB	77.0	78.3	80.1	82.2	85.7	88.8	89.4	89.0	86.7	83.6	80.0	79.1
	2%	WB	69.7	70.6	71.3	73.8	76.3	79.3	80.2	80.4	79.3	78.1	74.8	72.2
		MCDB	74.5	76.2	78.3	80.3	84.0	87.6	88.1	87.5	85.2	82.9	78.8	76.7
	5%	WB	68.0	69.1	69.6	72.4	75.3	78.2	79.3	79.6	78.4	76.9	72.9	70.7
		MCDB	72.9	74.6	76.3	79.1	82.6	86.1	87.2	86.7	84.3	82.0	76.9	75.0
	10%	WB	66.1	67.3	68.2	70.9	74.3	77.3	78.4	78.7	77.7	75.7	70.9	69.0
		MCDB	70.8	72.5	74.4	77.6	81.4	84.7	86.0	85.5	83.7	80.6	75.3	73.3
Mean Daily Temperature Range	5% DB	MDBR	19.5	19.3	19.1	18.8	17.6	15.6	15.5	14.6	13.2	14.7	17.1	17.8
		MCDBR	20.1	21.0	21.2	20.8	21.3	18.7	17.5	16.7	14.9	14.4	17.1	17.4
		MCWBR	11.4	10.7	9.7	8.8	7.8	7.0	6.6	6.4	5.8	6.3	8.9	9.6
	5% WB	MCDBR	17.8	18.1	18.2	17.8	17.1	16.6	16.0	14.8	13.0	12.1	14.0	15.5
		MCWBR	11.0	10.4	9.6	8.8	7.4	7.0	6.7	6.4	5.9	5.8	8.4	9.5
Clear-Sky Solar Irradiance	taub	0.341	0.354	0.372	0.394	0.429	0.458	0.483	0.471	0.425	0.396	0.360	0.348	
	taud	2.539	2.510	2.457	2.375	2.321	2.311	2.242	2.284	2.408	2.459	2.505	2.538	
	Ebn at Noon	285	290	291	286	275	265	258	261	271	272	275	277	
	Edh at Noon	28	31	35	39	41	42	44	42	36	32	29	27	
All-Sky Solar Radiation	RadAvg	1032	1245	1610	1948	2037	1892	1885	1781	1553	1373	1107	943	
	RadStd	77	103	127	104	137	173	106	116	112	100	64	76	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.72	N/A	N/A	N/A	N/A	N/A	-14	-86	+245	+202
(47)	Regional (37 neighbors)	+0.77	N/A	N/A	N/A	N/A	N/A	-11	-90	+274	+198

Nomenclature: See separate page